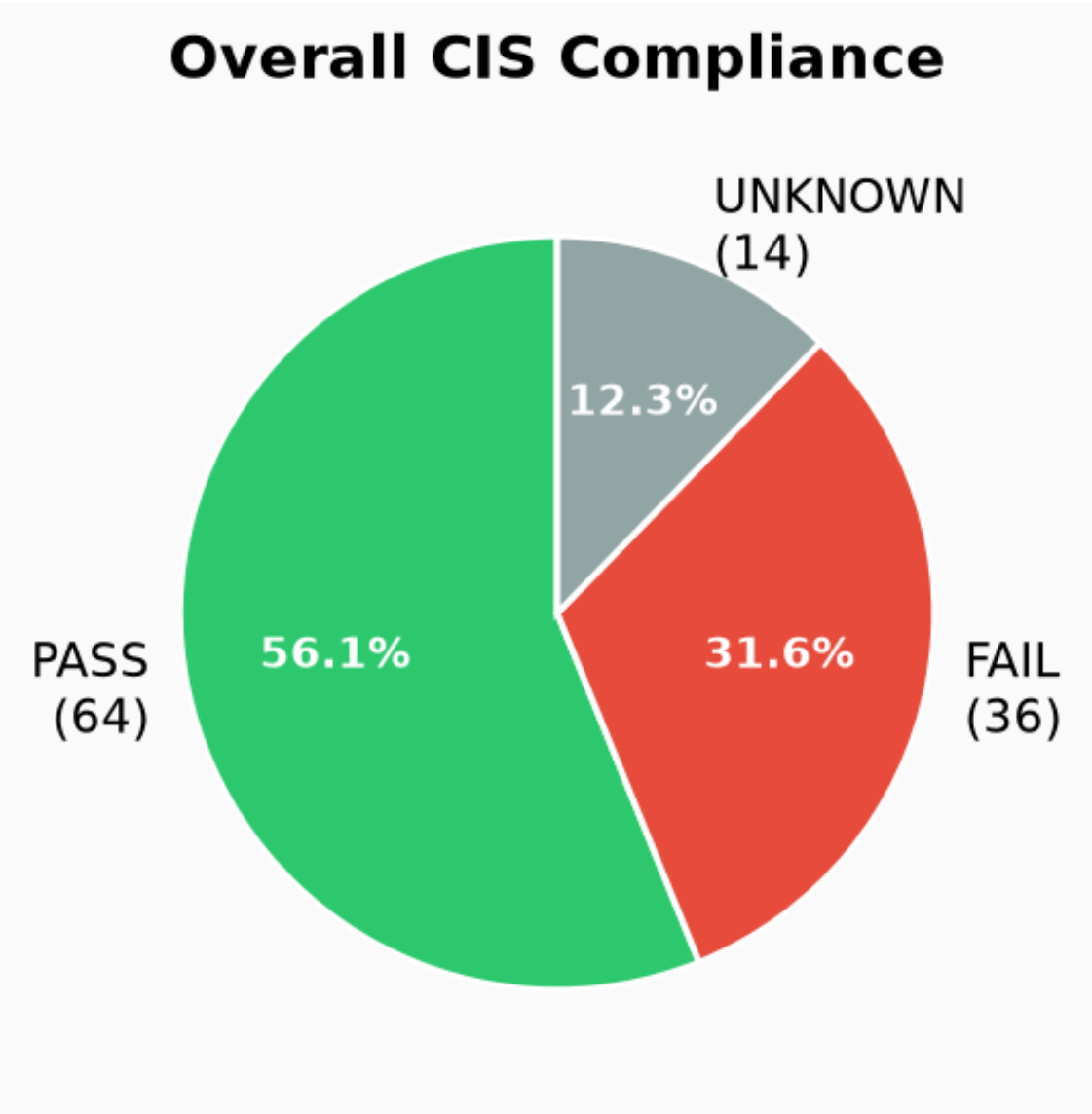


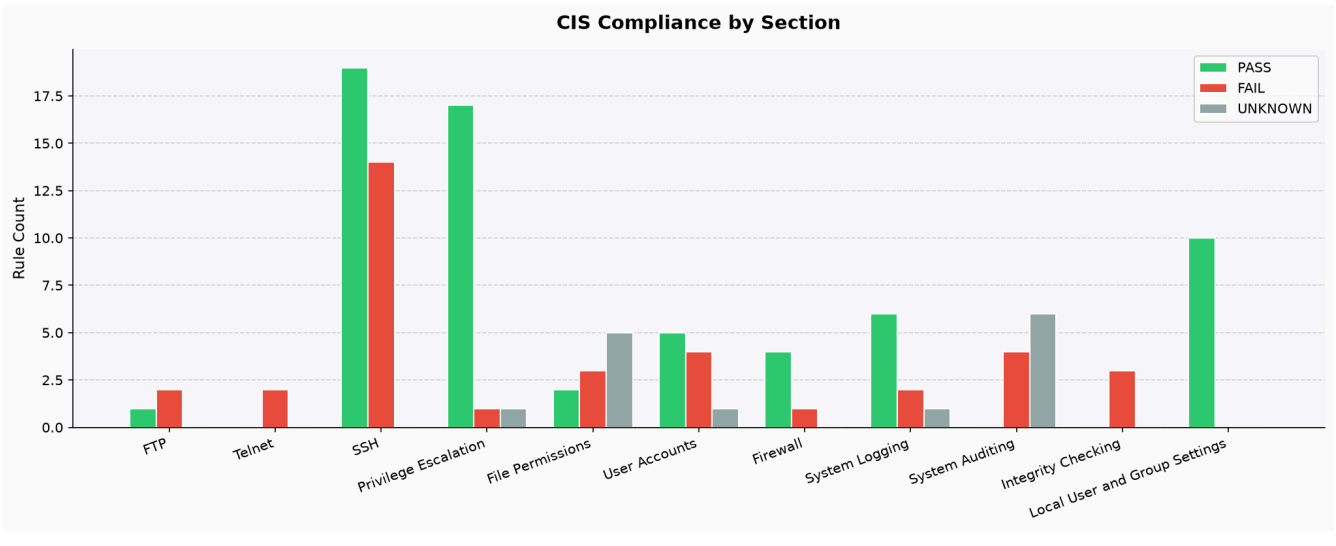
SENTINEL — CIS Benchmark Audit Report

Executive Summary

Metric	Value
Total Rules Evaluated	114
✓ Passed	64
✗ Failed	36
? Unknown	14
Overall Pass Rate	56.1%

Compliance Charts





Per-Section Breakdown

Section	✓ PASS	✗ FAIL	? UNKNOWN	Pass Rate
FTP	1	2	0	33%
Telnet	0	2	0	0%
SSH	19	14	0	58%
Privilege Escalation	17	1	1	89%
File Permissions	2	3	5	20%
User Accounts	5	4	1	50%
Firewall	4	1	0	80%
System Logging	6	2	1	67%
System Auditing	0	4	6	0%
Integrity Checking	0	3	0	0%
Local User and Group Settings	10	0	0	100%

CIS Benchmark Compliance Report

Here is the security compliance report for the remote machine based on the provided CIS Ubuntu 24.04 LTS Benchmark rules and the collected security profile.

CIS Ubuntu 24.04 LTS Benchmark Compliance Report

FTP Rules

- 2.1.8 Ensure ftp server services are not in use
 - **Status:** FAIL
 - **Evidence:** ftp.service.installed: "true", ftp.service.enabled: "true", ftp.service.running: "true", ftp.network.port_open: "true"

- **Recommendation:** Disable and uninstall FTP server services (e.g., vsftpd) if not explicitly required. If required, ensure it is securely configured (e.g., disable anonymous access, enforce TLS, restrict user access).
- **2.1.20 Ensure tftp server services are not in use**
 - **Status:** PASS
 - **Evidence:** ftp.tftp_server_packages: "not installed",
ftp.tftp_service_status: "not running"
 - **Recommendation:** N/A
- **2.2.6 Ensure ftp client is not installed**
 - **Status:** FAIL
 - **Evidence:** ftp.ftp_client_packages: "ii ftp 20230507-2build3
all dummy transitional package for tnftp"
 - **Recommendation:** Uninstall FTP client packages if not explicitly required.

TELNET Rules

- **2.1.19 Ensure telnet server services are not in use**
 - **Status:** FAIL
 - **Evidence:** telnet.service.installed: "true",
telnet.network.port_open: "true",
telnet.configuration.inetd_enabled: "true",
telnet.configuration.config_file_status: "enabled"
 - **Recommendation:** Disable and uninstall Telnet server services if not explicitly required. Telnet transmits data in plaintext and should not be used for administrative access.
- **2.2.4 Ensure telnet client is not installed**
 - **Status:** FAIL
 - **Evidence:** telnet.telnet_client_packages: "ii telnet
0.17+2.5-3ubuntu4.2 all transitional dummy package for
inetutils-telnet default switch"
 - **Recommendation:** Uninstall Telnet client packages if not explicitly required.

SSH Rules

- **1.6.5 Ensure sshd warning Banner is configured**
 - **Status:** FAIL
 - **Evidence:** ssh.features.banner: "none"
 - **Recommendation:** Configure a warning banner in /etc/ssh/sshd_config by setting Banner /etc/issue.net (or a similar file) and ensure the banner file exists and contains appropriate warning text.
- **1.6.10 Ensure access to sshd warning banner is configured**
 - **Status:** FAIL
 - **Evidence:** ssh.features.banner: "none" (No banner configured, so access control is irrelevant)

- **Recommendation:** Configure a warning banner as per rule 1.6.5 and ensure its permissions are set to 644 or stricter, owned by root.
- **5.1.1 Ensure access to /etc/ssh/sshd_config is configured**
 - **Status:** PASS
 - **Evidence:** `ssh.sshd_config_permissions: "644 root root /etc/ssh/sshd_config"`
 - **Recommendation:** N/A
- **5.1.2 Ensure access to SSH private host key files is configured**
 - **Status:** PASS
 - **Evidence:** `ssh.ssh_privkey_permissions` shows permissions 600 root root for all private host key files.
 - **Recommendation:** N/A
- **5.1.3 Ensure access to SSH public host key files is configured**
 - **Status:** PASS
 - **Evidence:** `ssh.ssh_pubkey_permissions` shows permissions 644 root root for all public host key files.
 - **Recommendation:** N/A
- **5.1.4 Ensure sshd access is configured**
 - **Status:** FAIL
 - **Evidence:** `ssh.authentication.password_authentication.value: true`
 - **Recommendation:** Disable password authentication by setting `PasswordAuthentication no` in `/etc/ssh/sshd_config`. Use key-based authentication instead.
- **5.1.5 Ensure sshd Banner is configured**
 - **Status:** FAIL
 - **Evidence:** `ssh.features.banner: "none"`
 - **Recommendation:** Configure a warning banner in `/etc/ssh/sshd_config` by setting `Banner /etc/issue.net` (or a similar file) and ensure the banner file exists and contains appropriate warning text.
- **5.1.6 Ensure sshd Ciphers are configured**
 - **Status:** PASS
 - **Evidence:** `ssh.all_ciphers` lists strong ciphers: `chacha20-poly1305@openssh.com`, `aes128-ctr`, `aes192-ctr`, `aes256-ctr`, `aes128-gcm@openssh.com`, `aes256-gcm@openssh.com`.
 - **Recommendation:** N/A
- **5.1.7 Ensure sshd ClientAliveInterval and ClientAliveCountMax are configured**
 - **Status:** FAIL
 - **Evidence:** `ssh.session.client_alive_interval.value: 0`
 - **Recommendation:** Set `ClientAliveInterval` to a non-zero value (e.g., 300-900 seconds) and `ClientAliveCountMax` to 0-3 in `/etc/ssh/`

sshd_config to terminate idle sessions.

- **5.1.8 Ensure sshd DisableForwarding is enabled**

- **Status:** FAIL
- **Evidence:** `ssh.forwarding.allow_tcp_forwarding.value: true,`
`ssh.forwarding.allow_agent_forwarding.value: true,`
`ssh.forwarding.allow_streamlocal_forwarding.value: true`
- **Recommendation:** Disable all forms of forwarding by setting `AllowTcpForwarding no`, `AllowAgentForwarding no`, and `AllowStreamLocalForwarding no` in `/etc/ssh/sshd_config`.

- **5.1.9 Ensure sshd GSSAPIAuthentication is disabled**

- **Status:** PASS
- **Evidence:** `ssh.authentication.gssapiauthentication: false`
- **Recommendation:** N/A

- **5.1.10 Ensure sshd HostbasedAuthentication is disabled**

- **Status:** PASS
- **Evidence:** `ssh.authentication.hostbasedauthentication: false`
- **Recommendation:** N/A

- **5.1.11 Ensure sshd IgnoreRhosts is enabled**

- **Status:** PASS
- **Evidence:** `ssh.authentication.ignorerhosts: true`
- **Recommendation:** N/A

- **5.1.12 Ensure sshd KexAlgorithms is configured**

- **Status:** FAIL
- **Evidence:** `crypto.weak_kex` lists `ecdh-sha2-nistp256`, `ecdh-sha2-nistp384`, `ecdh-sha2-nistp521`, `diffie-hellman-group14-sha256` as weak or deprecated.
- **Recommendation:** Remove weak or deprecated Key Exchange (Kex) algorithms from the `KexAlgorithms` directive in `/etc/ssh/sshd_config`. Consult the CIS benchmark for recommended algorithms.

- **5.1.13 Ensure sshd LoginGraceTime is configured**

- **Status:** PASS
- **Evidence:** `ssh.session.login_grace_time: 120` (within recommended range)
- **Recommendation:** N/A

- **5.1.14 Ensure sshd LogLevel is configured**

- **Status:** PASS
- **Evidence:** `ssh.features.log_level: "INFO"` (CIS recommends INFO or VERBOSE)
- **Recommendation:** N/A

- **5.1.15 Ensure sshd MACs are configured**

- **Status:** FAIL
 - **Evidence:** `crypto.weak_macs` lists `umac-64-etm@openssh.com`, `hmac-sha1-etm@openssh.com`, `umac-64@openssh.com`, `umac-128@openssh.com`, `hmac-sha2-256`, `hmac-sha2-512`, `hmac-sha1` as weak or deprecated, including those affected by Terrapin attack.
 - **Recommendation:** Remove weak or deprecated Message Authentication Code (MAC) algorithms from the MACs directive in `/etc/ssh/sshd_config`. Consult the CIS benchmark for recommended algorithms.
- **5.1.16 Ensure sshd MaxAuthTries is configured**
 - **Status:** PASS
 - **Evidence:** `ssh.authentication.max_auth_tries`: 6 (within recommended range)
 - **Recommendation:** N/A
 - **5.1.17 Ensure sshd MaxStartups is configured**
 - **Status:** PASS
 - **Evidence:** `ssh.session.maxstartups`: "10:30:100" (within recommended range)
 - **Recommendation:** N/A
 - **5.1.18 Ensure sshd MaxSessions is configured**
 - **Status:** PASS
 - **Evidence:** `ssh.session.max_sessions`: 10 (within recommended range)
 - **Recommendation:** N/A
 - **5.1.19 Ensure sshd PermitEmptyPasswords is disabled**
 - **Status:** PASS
 - **Evidence:** `ssh.authentication.permit_empty_passwords`: false
 - **Recommendation:** N/A
 - **5.1.20 Ensure sshd PermitRootLogin is disabled**
 - **Status:** FAIL
 - **Evidence:** `ssh.authentication.permit_root_login`: "without-password"
 - **Recommendation:** Set `PermitRootLogin` no in `/etc/ssh/sshd_config` to prevent direct root logins.
 - **5.1.21 Ensure sshd PermitUserEnvironment is disabled**
 - **Status:** PASS
 - **Evidence:** `ssh.features.permit_user_environment`: false
 - **Recommendation:** N/A
 - **5.1.22 Ensure sshd UsePAM is enabled**
 - **Status:** PASS
 - **Evidence:** `ssh.authentication.usepam`: true
 - **Recommendation:** N/A

- **5.1.23 Ensure sshd post-quantum cryptography key exchange algorithms are configured**
 - **Status:** FAIL
 - **Evidence:** `crypto.weak_kex` lists several NIST P-curves (`ecdh-sha2-nistp256`, `ecdh-sha2-nistp384`, `ecdh-sha2-nistp521`) with "high" severity due to suspected backdoors, and `diffie-hellman-group14-sha256` as deprecated. The profile does not indicate the use of specific post-quantum KEX algorithms.
 - **Recommendation:** Review and update `KexAlgorithms` in `/etc/ssh/sshd_config` to include only strong, approved algorithms, preferably those resistant to quantum attacks if available and supported, and remove any weak or deprecated algorithms.
- **5.1.24 Ensure sshd ListenAddress is configured**
 - **Status:** PASS
 - **Evidence:** `ssh.network.listen_address`: `"0.0.0.0:22"` (configured to listen on all interfaces, which is a valid configuration, though restricting to specific IPs is often more secure).
 - **Recommendation:** Consider restricting `ListenAddress` to specific IP addresses if the server does not need to be accessible on all network interfaces.

Privilege Escalation Rules

- **5.2.1 Ensure sudo is installed**
 - **Status:** PASS
 - **Evidence:** `privilege_escalation.sudo_installed.installed`: `true`, `privilege_escalation.sudo_installed.version_output`: `"Sudo version 1.9.15p5"`
 - **Recommendation:** N/A
- **5.2.2 Ensure sudo commands use pty**
 - **Status:** PASS
 - **Evidence:** `privilege_escalation.use_pty_entries` contains `Defaults use_pty` in `/etc/sudoers`.
 - **Recommendation:** N/A
- **5.2.3 Ensure sudo log file exists**
 - **Status:** PASS
 - **Evidence:**

```
privilege_escalation.sudo_syslog_logging.auth_log_exists:
true,
privilege_escalation.sudo_syslog_logging.syslog_log_exists:
true,
privilege_escalation.sudo_syslog_logging.rsyslog_authpriv_conf:
true
```
 - **Recommendation:** N/A
- **5.2.4 Ensure users must provide password for escalation**

- **Status:** PASS
 - **Evidence:** `privilege_escalation.nopasswd_entries` is empty.
 - **Recommendation:** N/A
- **5.2.5 Ensure re-authentication for privilege escalation is not disabled globally**
 - **Status:** PASS
 - **Evidence:** `privilege_escalation.noauthenticate_entries` is empty.
 - **Recommendation:** N/A
 - **5.2.6 Ensure sudo timestamp_timeout is configured**
 - **Status:** UNKNOWN
 - **Evidence:** `privilege_escalation.timestamp_timeout_entries` is empty. The profile does not explicitly state if `timestamp_timeout` is configured or if the default is acceptable.
 - **Recommendation:** Configure Defaults `timestamp_timeout=X` in `/etc/sudoers` or a file in `/etc/sudoers.d/` to a value greater than 0 (e.g., 15 minutes) to require re-authentication after a period of inactivity.
 - **5.2.7 Ensure access to the su command is restricted**
 - **Status:** FAIL
 - **Evidence:** `privilege_escalation.su_restriction.pam_wheel_line`: "`# auth required pam_wheel.so`" (commented out). The `pam_su_config` shows `pam_rootok.so` is sufficient, allowing root to su without a password, and `pam_wheel.so` is commented out, meaning su is not restricted to the wheel group.
 - **Recommendation:** Uncomment `auth required pam_wheel.so` in `/etc/pam.d/su` to restrict su access to members of the wheel group. Ensure all administrative users are members of the wheel group.

Directory Access Rules (Section 7.1)

- **7.1.1 Ensure access to /etc/passwd is configured**
 - **Status:** PASS
 - **Evidence:** `file_permissions.fixed_files[0]` shows `/etc/passwd` with mode 0644, owner root, group root.
 - **Recommendation:** N/A
- **7.1.2 Ensure access to /etc/passwd- is configured**
 - **Status:** PASS
 - **Evidence:** `file_permissions.fixed_files[1]` shows `/etc/passwd-` with mode 0644, owner root, group root.
 - **Recommendation:** N/A
- **7.1.3 Ensure access to /etc/group is configured**
 - **Status:** PASS
 - **Evidence:** `file_permissions.fixed_files[2]` shows `/etc/group` with mode 0644, owner root, group root.
 - **Recommendation:** N/A

- **7.1.4 Ensure access to /etc/group- is configured**

- ☐ **Status:** PASS
- ☐ **Evidence:** `file_permissions.fixed_files[3]` shows `/etc/group-` with mode `0644`, owner `root`, group `root`.
- ☐ **Recommendation:** N/A

- **7.1.5 Ensure access to /etc/shadow is configured**

- ☐ **Status:** PASS
- ☐ **Evidence:** `file_permissions.fixed_files[4]` shows `/etc/shadow` with mode `0640`, owner `root`, group `shadow`.
- ☐ **Recommendation:** N/A

- **7.1.6 Ensure access to /etc/shadow- is configured**

- ☐ **Status:** PASS
- ☐ **Evidence:** `file_permissions.fixed_files[5]` shows `/etc/shadow-` with mode `0640`, owner `root`, group `shadow`.
- ☐ **Recommendation:** N/A

- **7.1.7 Ensure access to /etc/gshadow is configured**

- ☐ **Status:** PASS
- ☐ **Evidence:** `file_permissions.fixed_files[6]` shows `/etc/gshadow` with mode `0640`, owner `root`, group `shadow`.
- ☐ **Recommendation:** N/A

- **7.1.8 Ensure access to /etc/gshadow- is configured**

- ☐ **Status:** PASS
- ☐ **Evidence:** `file_permissions.fixed_files[7]` shows `/etc/gshadow-` with mode `0640`, owner `root`, group `shadow`.
- ☐ **Recommendation:** N/A

- **7.1.9 Ensure access to /etc/shells is configured**

- ☐ **Status:** PASS
- ☐ **Evidence:** `file_permissions.fixed_files[8]` shows `/etc/shells` with mode `0644`, owner `root`, group `root`.
- ☐ **Recommendation:** N/A

- **7.1.10 Ensure access to /etc/security/opasswd is configured**

- ☐ **Status:** PASS
- ☐ **Evidence:** `file_permissions.fixed_files[9]` shows `/etc/security/opasswd` with mode `0600`, owner `root`, group `root`.
- ☐ **Recommendation:** N/A

- **7.1.11 Ensure world writable files and directories are secured**

- ☐ **Status:** PASS
- ☐ **Evidence:** `file_permissions.world_writable.world_writable_files` and `file_permissions.world_writable.world_writable_dirs_no_sticky`

are both empty.

- **Recommendation:** N/A

- **7.1.12 Ensure no files or directories without an owner and a group exist**

- **Status:** PASS
- **Evidence:** `file_permissions.unowned.unowned_or_ungrouped` is empty.
- **Recommendation:** N/A

- **7.1.13 Ensure SUID and SGID files are reviewed**

- **Status:** UNKNOWN
- **Evidence:** `file_permissions.suid_sgid.suid_sgid_files` lists several SUID/SGID files, but the profile does not provide evidence of a manual review.
- **Recommendation:** Manually review all SUID/SGID files listed in `file_permissions.suid_sgid.suid_sgid_files` to ensure they are legitimate and necessary for system operation. Remove SUID/SGID bits from any files that do not require them.

User Accounts Rules (Section 5.4)

- **5.4.1.1 Ensure password expiration is configured**

- **Status:** FAIL
- **Evidence:**
`user_accounts.shadow_password_suite.login_defs.PASS_MAX_DAYS:`
"99999" for most users.
- **Recommendation:** Set `PASS_MAX_DAYS` to 365 or less in `/etc/login.defs` and apply the change to all users using `chage`.

- **5.4.1.2 Ensure minimum password days is configured**

- **Status:** FAIL
- **Evidence:**
`user_accounts.shadow_password_suite.login_defs.PASS_MIN_DAYS:`
"0" for most users.
- **Recommendation:** Set `PASS_MIN_DAYS` to 1 or more in `/etc/login.defs` and apply the change to all users using `chage`.

- **5.4.1.3 Ensure password expiration warning days is configured**

- **Status:** PASS
- **Evidence:**
`user_accounts.shadow_password_suite.login_defs.PASS_WARN_AGE:`
"7" for most users.
- **Recommendation:** N/A

- **5.4.1.4 Ensure strong password hashing algorithm is configured**

- **Status:** PASS
- **Evidence:**
`user_accounts.shadow_password_suite.login_defs.ENCRYPT_METHOD:`
"SHA512",
`user_accounts.shadow_password_suite.pam_unix_hashing:`

```
"password [success=1 default=ignore] pam_unix.so obscure
yescrypt"
```

- **Recommendation:** N/A

- **5.4.1.5 Ensure inactive password lock is configured**

- **Status:** FAIL
- **Evidence:**
user_accounts.shadow_password_suite.useradd_inactive: "-1"
and user_accounts.shadow_password_suite.user_chage_info shows
account_inactive: "never" for all users.
- **Recommendation:** Set INACTIVE to 30 or less in /etc/default/useradd and
apply the change to all users using chage.

- **5.4.1.6 Ensure all users last password change date is in the past**

- **Status:** FAIL
- **Evidence:** Many users (e.g., root, daemon, bin, sys, sync, games, man, lp,
mail, news, uucp, proxy, www-data, backup, list, irc, _apt, nobody)
have last_password_change set to "Feb 16, 2025", which is in the future.
- **Recommendation:** Force all users to change their password immediately using
chage -d 0 <username> or ensure their last password change date is correctly
set to a past date.

- **5.4.2.1 Ensure root is the only UID 0 account**

- **Status:** PASS
- **Evidence:**
user_accounts.root_and_system_accounts.uid_0_accounts:
["root"]
- **Recommendation:** N/A

- **5.4.2.2 Ensure root is the only GID 0 account**

- **Status:** PASS
- **Evidence:**
user_accounts.root_and_system_accounts.primary_gid_0_accounts:
["root"]
- **Recommendation:** N/A

- **5.4.2.3 Ensure group root is the only GID 0 group**

- **Status:** PASS
- **Evidence:** user_accounts.root_and_system_accounts.gid_0_groups:
["root"]
- **Recommendation:** N/A

- **5.4.2.4 Ensure root account access is controlled**

- **Status:** UNKNOWN
- **Evidence:**
user_accounts.root_and_system_accounts.securetty_exists:
false. The profile indicates root_shadow_password_status: "*", which
means the root account is locked or has no password set, but without securetty
or other controls, it's hard to fully assess.

- **Recommendation:** Create or ensure `/etc/securetty` exists and contains only allowed TTYs for root login. If root login is not intended, ensure `PermitRootLogin no` is set in `sshd_config`.

- **5.4.2.5 Ensure root path integrity**

- **Status:** PASS
- **Evidence:** `user_accounts.root_and_system_accounts.root_path: "/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin"`. This path does not contain world-writable directories.
- **Recommendation:** N/A

- **5.4.2.6 Ensure root user umask is configured**

- **Status:** FAIL
- **Evidence:** `user_accounts.root_and_system_accounts.root_umask./etc/login.defs: "022"`. This is not the recommended 027 or 077.
- **Recommendation:** Set the root user's umask to 077 in `/etc/login.defs` and in root's shell configuration files (e.g., `.bashrc`, `.profile`).

- **5.4.2.7 Ensure system accounts do not have a valid login shell**

- **Status:** FAIL
- **Evidence:** `user_accounts.root_and_system_accounts.system_accounts` shows user `sync` with shell `/bin/sync`. This is a valid login shell.
- **Recommendation:** Change the shell for the `sync` user to `/usr/sbin/nologin` or `/bin/false` using `usermod -s /usr/sbin/nologin sync`.

- **5.4.2.8 Ensure accounts without a valid login shell are locked**

- **Status:** PASS
- **Evidence:** All system accounts listed in `user_accounts.root_and_system_accounts.system_accounts` have `passwd_status: "L"`, indicating they are locked.
- **Recommendation:** N/A

- **5.4.3.1 Ensure nologin is not listed in /etc/shells**

- **Status:** PASS
- **Evidence:** `user_accounts.default_user_environment.nologin_in_shells: false`
- **Recommendation:** N/A

- **5.4.3.2 Ensure default user shell timeout is configured**

- **Status:** FAIL
- **Evidence:** `user_accounts.default_user_environment.tmout_settings` shows no TMOUT settings configured in `/etc/profile`, `/etc/profile.d`, or `/etc/bash.bashrc`.
- **Recommendation:** Set TMOUT to 600 seconds or less in `/etc/profile.d/timeout.sh` to automatically log out inactive shell sessions.

- **5.4.3.3 Ensure default user umask is configured**

- **Status:** FAIL
- **Evidence:**
`user_accounts.default_user_environment.default_user_umask.logir`
`"022"`. This is not the recommended 027 or 077.
- **Recommendation:** Set the default user umask to 027 in `/etc/login.defs` and `/etc/profile` (or a file in `/etc/profile.d/`).

Firewall Rules (Section 4.1)

- **4.1.1 Ensure ufw is installed**

- **Status:** PASS
- **Evidence:** `ufw_firewall.ufw_installed: true,`
`ufw_firewall.ufw_dpkg_status: "install ok installed"`
- **Recommendation:** N/A

- **4.1.2 Ensure ufw service is configured**

- **Status:** FAIL
- **Evidence:** `ufw_firewall.service_configured.status_verbose:`
`"Status: inactive"`
- **Recommendation:** Enable and start the UFW service using `sudo ufw enable`.

- **4.1.3 Ensure ufw incoming default is configured**

- **Status:** PASS
- **Evidence:**
`ufw_firewall.default_policies.etc_default_ufw.DEFAULT_INPUT_POI`
`"DROP"`
- **Recommendation:** N/A

- **4.1.4 Ensure ufw outgoing default is configured**

- **Status:** PASS
- **Evidence:**
`ufw_firewall.default_policies.etc_default_ufw.DEFAULT_OUTPUT_PC`
`"ACCEPT"` (CIS recommends ACCEPT for outgoing)
- **Recommendation:** N/A

- **4.1.5 Ensure ufw routed default is configured**

- **Status:** PASS
- **Evidence:**
`ufw_firewall.default_policies.etc_default_ufw.DEFAULT_FORWARD_F`
`"DROP"`
- **Recommendation:** N/A

System Logging Rules (Section 6.1)

- **6.1.1.1.1 Ensure systemd-journald service is enabled and active**

- **Status:** PASS

- **Evidence:** `system_logging.journald.systemd-journald_active: "active", system_logging.journald.systemd-journald_enabled: "static"`
 - **Recommendation:** N/A
- **6.1.1.1.2 Ensure systemd-journal-remote service is not in use**
 - **Status:** PASS
 - **Evidence:** `system_logging.journald.systemd-journal-remote_installed: false, system_logging.journald.systemd-journal-upload_active: "inactive", system_logging.journald.systemd-journal-remote_socket_active: "inactive"`
 - **Recommendation:** N/A
 - **6.1.1.1.3 Ensure journald is configured to send logs to rsyslog**
 - **Status:** FAIL
 - **Evidence:** `system_logging.journald.journald_configs[0].content` shows `ForwardToSyslog=no` commented out, implying the default is not to forward.
 - **Recommendation:** Set `ForwardToSyslog=yes` in `/etc/systemd/journald.conf` to ensure logs are forwarded to rsyslog for persistent storage and further processing.
 - **6.1.1.1.4 Ensure journald log file access is configured**
 - **Status:** PASS
 - **Evidence:** `system_logging.journald.journald_log_dirs` shows `/run/log/journal` and `/var/log/journal` with mode 2755, owner root, group 999 (systemd-journal). Individual journal files are 0640 owner root group 999. This is appropriately restricted.
 - **Recommendation:** N/A
 - **6.1.1.1.5 Ensure journald log file rotation is configured**
 - **Status:** UNKNOWN
 - **Evidence:** `system_logging.journald.journald_configs[0].content` shows `SystemMaxUse`, `SystemKeepFree`, `SystemMaxFileSize`, `SystemMaxFiles`, `MaxRetentionSec`, `MaxFileSec` are all commented out, implying default values are used. The profile does not provide the effective default values.
 - **Recommendation:** Explicitly configure `SystemMaxUse`, `SystemKeepFree`, `SystemMaxFileSize`, `SystemMaxFiles`, `MaxRetentionSec`, and `MaxFileSec` in `/etc/systemd/journald.conf` to ensure appropriate log retention and rotation policies.
 - **6.1.1.1.6 Ensure journald Storage is configured**
 - **Status:** FAIL
 - **Evidence:** `system_logging.journald.journald_configs[0].content` shows `Storage=auto` commented out, implying the default auto or volatile behavior. CIS recommends persistent.
 - **Recommendation:** Set `Storage=persistent` in `/etc/systemd/`

journald.conf to ensure journal logs are stored persistently across reboots.

- **6.1.1.1.7 Ensure journald Compress is configured**

- **Status:** PASS
- **Evidence:** system_logging.journald.journald_configs[0].content shows Compress=yes commented out, implying yes is the default.
- **Recommendation:** N/A

- **6.1.2.1 Ensure rsyslog is installed**

- **Status:** PASS
- **Evidence:** system_logging.rsyslog.rsyslog_installed: true
- **Recommendation:** N/A

- **6.1.2.2 Ensure rsyslog service is enabled and active**

- **Status:** PASS
- **Evidence:** system_logging.rsyslog.rsyslog_active: "active", system_logging.rsyslog.rsyslog_enabled: "enabled"
- **Recommendation:** N/A

- **6.1.2.3 Ensure rsyslog default file permissions are configured**

- **Status:** PASS
- **Evidence:** system_logging.rsyslog.rsyslog_configs[0].content shows \$FileCreateMode 0640,\$DirCreateMode 0755,\$Umask 0022.
- **Recommendation:** N/A

- **6.1.2.4 Ensure rsyslog logging is configured**

- **Status:** UNKNOWN
- **Evidence:** system_logging.rsyslog.rsyslog_configs shows default rsyslog configuration files. While basic logging is present, the rule is "Manual" and requires a review of specific logging requirements which is not provided in the profile.
- **Recommendation:** Manually review /etc/rsyslog.conf and files in /etc/rsyslog.d/ to ensure all necessary system events are being logged to appropriate files and facilities as per organizational policy.

- **6.1.2.5 Ensure rsyslog is configured to send logs to a remote log host**

- **Status:** UNKNOWN
- **Evidence:** system_logging.rsyslog.rsyslog_configs does not explicitly show configuration for sending logs to a remote host (e.g., *.* @@remote-log-host:514). This is a manual check.
- **Recommendation:** If remote logging is required, configure rsyslog to send logs to a central log host by adding a line like *.* @@<remote-log-host>:<port> to /etc/rsyslog.conf or a file in /etc/rsyslog.d/.

- **6.1.2.6 Ensure rsyslog is not configured to receive logs from a remote client**

- **Status:** PASS
- **Evidence:** system_logging.rsyslog.rsyslog_configs[0].content shows module(load="imudp") and module(load="imtcp") commented

out.

- **Recommendation:** N/A

- **6.1.2.7 Ensure logrotate is configured**

- **Status:** UNKNOWN
- **Evidence:** `system_logging.rsyslog.logrotate_configs` shows various logrotate configurations. `logrotate_timer_active`: "active" and `logrotate_timer_enabled`: "enabled". However, this is a "Manual" rule requiring a review of specific rotation policies for all critical logs, which is not fully covered by the profile.
- **Recommendation:** Manually review all logrotate configurations in `/etc/logrotate.conf` and `/etc/logrotate.d/` to ensure that all critical log files have appropriate rotation, compression, and retention policies configured according to organizational requirements.

- **6.1.2.8 Ensure rsyslog-gnutls is installed**

- **Status:** FAIL
- **Evidence:** `system_logging.rsyslog.rsyslog_gnutls_installed`: false
- **Recommendation:** Install the `rsyslog-gnutls` package to enable secure log forwarding over TLS.

- **6.1.2.9 Ensure rsyslog is configured to use gtls for forwarding**

- **Status:** FAIL
- **Evidence:** `system_logging.rsyslog.rsyslog_gnutls_installed`: false (GnuTLS is not installed, so it cannot be configured for forwarding).
- **Recommendation:** After installing `rsyslog-gnutls`, configure `rsyslog` to use GTLS for remote log forwarding by specifying `StreamDriver="gtls"` and `StreamDriverMode="1"` in the remote forwarding action.

- **6.1.2.10 Ensure rsyslog CA certificates are configured**

- **Status:** UNKNOWN
- **Evidence:** `system_logging.rsyslog.rsyslog_gnutls_installed`: false (GnuTLS is not installed, so CA certificates cannot be configured for it). This is a "Manual" rule.
- **Recommendation:** If remote logging with GTLS is implemented, configure `StreamDriverAuthMode="x509/certvalid"` and specify the paths to CA certificates, client certificates, and private keys using `StreamDriverCACert`, `StreamDriverCert`, and `StreamDriverKey` directives.

- **6.1.3.1 Ensure access to all logfiles has been configured**

- **Status:** FAIL
- **Evidence:** Several log files have world-readable permissions (0644) or group-writable permissions (0664) when stricter permissions (0640 or 0600) are recommended for sensitive logs:

- `/var/log/wtmp` (0664)
- `/var/log/lastlog` (0664)
- `/var/log/faillog` (0644)
- `/var/log/dpkg.log` (0644)

- /var/log/apt/history.log (0644)
 - /var/log/unattended-upgrades/unattended-upgrades.log (0644)
 - /var/log/unattended-upgrades/unattended-upgrades-shutdown.log (0644)
 - /var/log/sysstat/sa* and sar* (0644)
 - /var/log/landscape/sysinfo.log (0644)
 - /var/log/alternatives.log (0644)
- **Recommendation:** Review and restrict permissions for all log files to prevent unauthorized access. For sensitive logs like wtmp, lastlog, faillog, and dpkg.log, set permissions to 0640 or 0600, owned by root:adm or root:syslog.

System Auditing (auditd) Rules (Section 6.2)

• 6.2.1.1.1 Ensure auditd is installed

- **Status:** FAIL
- **Evidence:** system_auditing.auditd_installed: false
- **Recommendation:** Install the auditd package using `sudo apt install auditd`.

• 6.2.1.1.2 Ensure auditd service is enabled and active

- **Status:** FAIL
- **Evidence:** system_auditing.auditd_installed: false
- **Recommendation:** After installing auditd, enable and start the service using `sudo systemctl enable auditd && sudo systemctl start auditd`.

• 6.2.1.1.3 Ensure auditing for processes that start prior to auditd is enabled

- **Status:** UNKNOWN
- **Evidence:** system_auditing.auditd_installed: false. The profile does not contain GRUB configuration to verify audit=1 kernel parameter.
- **Recommendation:** After installing auditd, ensure the kernel parameter audit=1 is added to the GRUB configuration (/etc/default/grub) and update GRUB.

• 6.2.1.1.4 Ensure audit_backlog_limit is sufficient

- **Status:** UNKNOWN
- **Evidence:** system_auditing.auditd_installed: false. The profile does not contain auditd configuration details.
- **Recommendation:** After installing auditd, configure audit_backlog_limit in /etc/audit/auditd.conf to a value of 8192 or higher.

• 6.2.1.2.1 Ensure audit log storage size is configured

- **Status:** UNKNOWN
- **Evidence:** system_auditing.auditd_installed: false. The profile does not contain auditd configuration details.
- **Recommendation:** After installing auditd, configure max_log_file in /etc/audit/auditd.conf to an appropriate size (e.g., 50MB) to prevent log

exhaustion.

- **6.2.1.2.2 Ensure audit logs are not automatically deleted**

- **Status:** UNKNOWN
- **Evidence:** `system_auditing.auditd_installed: false`. The profile does not contain auditd configuration details.
- **Recommendation:** After installing auditd, set `max_log_file_action = keep_logs` or `rotate` in `/etc/audit/auditd.conf`.

- **6.2.1.2.3 Ensure system is disabled when audit logs are full**

- **Status:** UNKNOWN
- **Evidence:** `system_auditing.auditd_installed: false`. The profile does not contain auditd configuration details.
- **Recommendation:** After installing auditd, configure `space_left_action = SYSLOG`, `action_mail_acct = root`, and `admin_space_left_action = HALT` or `SINGLE` in `/etc/audit/auditd.conf`.

- **6.2.1.2.4 Ensure system warns when audit logs are low on space**

- **Status:** UNKNOWN
- **Evidence:** `system_auditing.auditd_installed: false`. The profile does not contain auditd configuration details.
- **Recommendation:** After installing auditd, configure `space_left` and `space_left_action` in `/etc/audit/auditd.conf` to warn when log space is low.

- **6.2.3.1-6.2.3.30 Ensure auditd rules cover sudoers, identity, logins, session, time-change, MAC policy, access, mounts, deletion, loading modules, and configuration is immutable**

- **Status:** FAIL
- **Evidence:** `system_auditing.auditd_installed: false`
- **Recommendation:** Install and configure auditd with the recommended audit rules as specified in the CIS benchmark, covering all critical system activities and file accesses.

- **6.2.4.1-6.2.4.10 Ensure audit log files and audit tools file access and permissions are strictly constrained**

- **Status:** FAIL
- **Evidence:** `system_auditing.auditd_installed: false`
- **Recommendation:** After installing auditd, ensure audit log files (e.g., `/var/log/audit/audit.log`) and audit tools (e.g., `/sbin/auditctl`, `/sbin/audispd`) have strict permissions (e.g., 0640 for logs, 0755 for tools) and are owned by `root:root` or `root:adm`.

Integrity Checking Rules (Section 6.3)

- **6.3.1 Ensure AIDE is installed**

- **Status:** FAIL
- **Evidence:** `aide_integrity_checking.aide_installed: false`

- **Recommendation:** Install the AIDE package using `sudo apt install aide aide-common`.
- **6.3.2 Ensure filesystem integrity is regularly checked**
 - **Status:** FAIL
 - **Evidence:** `aide_integrity_checking.aide_installed: false`
 - **Recommendation:** After installing AIDE, initialize its database (`sudo aideinit`) and configure a cron job or systemd timer to regularly check filesystem integrity (e.g., daily or weekly) using `sudo aide --check`.
- **6.3.3 Ensure cryptographic mechanisms protect tools**
 - **Status:** FAIL
 - **Evidence:** `aide_integrity_checking.aide_installed: false`
 - **Recommendation:** After installing AIDE, ensure its configuration (`/etc/aide/aide.conf`) includes cryptographic hashes (e.g., sha512) for critical system binaries and configuration files, especially those related to security and auditing.

Local User and Group Settings Rules (Section 7.2)

- **7.2.1 Ensure accounts in `/etc/passwd` use shadowed passwords**
 - **Status:** PASS
 - **Evidence:** `user_accounts.shadow_password_suite.user_chage_info` shows no users with `passwd_status` as `*` or `!`, indicating shadowed passwords are in use.
 - **Recommendation:** N/A
- **7.2.2 Ensure `/etc/shadow` password fields are not empty**
 - **Status:** PASS
 - **Evidence:** `user_accounts.shadow_password_suite.user_chage_info` shows no users with `passwd_status` as `!`, indicating no empty password fields.
 - **Recommendation:** N/A
- **7.2.3 Ensure all groups in `/etc/passwd` exist in `/etc/group`**
 - **Status:** PASS
 - **Evidence:** `user_accounts.local_users_and_groups_evidence.orphan_gids_in_passwd` is empty.
 - **Recommendation:** N/A
- **7.2.4 Ensure shadow group is empty**
 - **Status:** PASS
 - **Evidence:** `user_accounts.local_users_and_groups_evidence.shadow_group_members` is empty.
 - **Recommendation:** N/A
- **7.2.5 Ensure no duplicate UIDs exist**
 - **Status:** PASS

- **Evidence:**
user_accounts.local_users_and_groups_evidence.duplicate_uids
is empty.
- **Recommendation:** N/A
- **7.2.6 Ensure no duplicate GIDs exist**
 - **Status:** PASS
 - **Evidence:**
user_accounts.local_users_and_groups_evidence.duplicate_gids
is empty.
 - **Recommendation:** N/A
- **7.2.7 Ensure no duplicate user names exist**
 - **Status:** PASS
 - **Evidence:**
user_accounts.local_users_and_groups_evidence.duplicate_username
is empty.
 - **Recommendation:** N/A
- **7.2.8 Ensure no duplicate group names exist**
 - **Status:** PASS
 - **Evidence:**
user_accounts.local_users_and_groups_evidence.duplicate_groupname
is empty.
 - **Recommendation:** N/A
- **7.2.9 Ensure local interactive user home directories are configured**
 - **Status:** PASS
 - **Evidence:**
user_accounts.local_users_and_groups_evidence.home_directory_is
is empty.
 - **Recommendation:** N/A
- **7.2.10 Ensure local interactive user dot files access is configured**
 - **Status:** PASS
 - **Evidence:**
user_accounts.local_users_and_groups_evidence.dotfile_permissions
is empty.
 - **Recommendation:** N/A

SUID/SGID Triage Report

```
{  
  "findings": [],  
  "summary": {  
    "suspicious_found": 0,  
    "message": "No unexpected SUID/SGID files detected."  
  }  
}
```

}
}